Ranking Within Fixed Source Allocations: Controlled Mathematical Instruction Selection

Xin Huang*, Tianle Xu†, Changjun Ren‡, Yuhuang Chen§,
Heng Zhan¶, Jiawen Huang*, Shijie Jia*
*Dalian Jiaotong University, Dalian, China
†Fudan University, Shanghai, China
‡Shanghai University of Engineering Science, Shanghai, China
§Xi'an Jiaotong-Liverpool University, Suzhou, China
¶Harbin Institute of Technology, Harbin, China
20252340@djtu.edu.cn, xutianle@fudan.edu.cn, m325125218@sues.edu.cn
Yuhuang.Chen22@student.xjtlu.edu.cn, zhanheng288@gmail.com
wodenaozi@gmail.com, jsj@djtu.edu.cn

***Abstract*—Generated instruction pools contain multiple records derived from the same source question, making data selection a joint decision about source allocation and record ranking. Global comparisons combine these decisions, while equal input-token budgets can still yield different amounts of supervised training. We introduce a source-controlled evaluation protocol that compares ranked and uniformly sampled records within exact source, type, and token-length cells. The paired selections match source quotas, record counts, input tokens, supervised tokens, and training updates. Using 223,553 MetaMathQA records and Qwen2.5-1.5B, we evaluate a LESS-based ranking across three subset constructions and three training seeds under a shared scorer. In the retrospective dev256 comparison, mean strict reference matching reaches 79.731%, compared with 78.429% for matched random selection, a relative increase of 1.66%. The mean difference is positive for every construction and remains positive after omitting any one construction or training seed. The conditional 95% question-bootstrap interval for the absolute difference is $[0.000, 2.604]$ percentage points. The protocol makes ranking value measurable under fixed allocations and shows why source quotas and realized supervision should accompany comparisons of instruction-selection methods.**

***Index Terms*—instruction tuning, data selection, large language models, source allocation, controlled evaluation**

I. Introduction

Instruction-data selection is a resource-allocation decision: which problems receive training effort, and which examples represent each problem? Instruction-generation methods create synthetic data [1]; in MetaMathQA, retained original-question fields group alternative phrasings, solutions, and task variants. Under a fixed budget, source allocation, coverage, and record value describe different properties of the training collection.

Target-guided methods estimate record relevance from parameter sensitivity, training gradients, or related signals [2]–[11]. A global comparison can change source allocation and within-source ranking together, while equal input-token ceilings still allow supervised-token exposure and update counts to differ. This creates an attribution problem: a downstream difference can reflect which source questions receive training effort, which records represent them, or how much supervision the model receives.

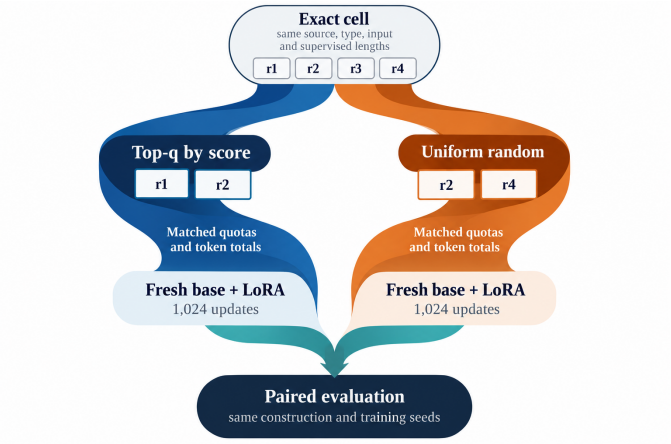


Fig. 1. Within-source selection under matched allocations. Each exact source–type–input/supervised-length cell supplies top-q and uniform-random arms with matched quotas and token totals. Both use fresh LoRA fits and paired construction/training seeds; arms may overlap.

We introduce a source-controlled evaluation protocol that pairs ranked and uniformly sampled records within exact source–type–length cells. Shared cell quotas preserve source identities, allocations, record counts, both token totals, and training updates (Fig. 1), making ranking value measurable within a fixed allocation.

Using MetaMathQA [12], GSM8K [13], Qwen2.5-1.5B [14], and a 2M input-token ceiling, our paired selections cover approximately 63% of pool sources. The retrospective dev256 comparison shows a 1.66% relative increase in mean strict reference matching for the LESS adaptation, with positive averages across all three constructions. Global comparisons reveal different supervised-token totals under nearly identical input budgets. Together, these findings motivate recording source allocation and realized supervision alongside ranking performance. Auxiliary diagnostics examine source coverage and evaluation-reference validity.

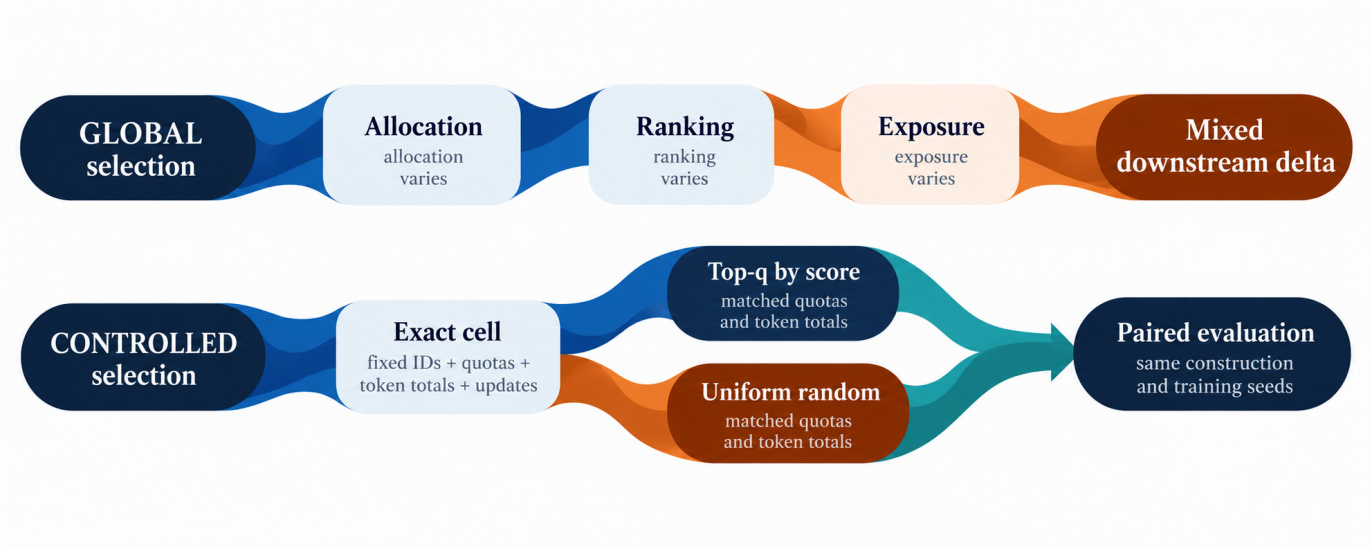


Fig. 2. Global and controlled selection comparisons. The global path varies allocation, ranking, and exposure jointly. The controlled path fixes source IDs, quotas, token totals, and updates before comparing top-q with uniform random and paired evaluation.

II. Related Work

Target relevance. LESS combines LoRA warmup, candidate Adam update directions, projected cosine similarity to target gradients, and checkpoint aggregation [4]. GIST uses target-gradient subspaces [6], while Influence Distillation approximates candidate–target relationships through embeddings [15]. Our protocol evaluates relevance-guided ranking under explicit source allocations. We instantiate it with a LESS adaptation and separately examine an initialization-gradient diagnostic.

Coverage, redundancy, and budgets. Representation, gradient clustering, diversity, deduplication, and quality–diversity methods change what a fixed record budget represents [5], [7], [8], [16]–[19]. Training-dynamics diagnostics provide another view of example-level variability and dataset structure [20]. Compute-Constrained Data Selection and importance resampling study token budgets and target matching [9], [11]; LESS also discusses response-length and step-alignment effects [4]. Our control fixes source IDs and quotas, then matches types, exact lengths, input tokens, supervised tokens, and updates within substitution cells. This defines a conditional comparison of ranking rules within the same source allocation.

III. Controlled Selection Protocol

A. Source Mapping and Evaluation Sets

We normalize MetaMathQA `original_question` fields using NFKC and whitespace normalization and retain unique lexical matches to GSM8K train questions. Of 395,000 MetaMathQA records, 240,000 map uniquely to 7,390 of the 7,473 GSM8K train questions. Removing records from the source-disjoint probe256 and dev256 partitions leaves 223,553 candidates from 6,878 sources. The original collection includes 8,438 probe-source and 8,009 dev-source derivatives; these are excluded from the pool. The other 155,000 records lack this unique mapping and are also excluded.

Probe256 supplies labeled target information for scoring. Dev256 contains 256 held-out GSM8K train questions. Secondary83 contains the 83 train questions outside the 7,390 uniquely mapped sources and is selected by mapping absence. The pool and all three evaluation-related sets have pairwise disjoint source IDs and normalized queries. Historical checks of selection manifests, training orders, and update blocks recorded no probe, dev, or secondary source among the 51 downstream fits; the member-level records needed to repeat those checks were not retained. Of these fits, 39 use probe-conditioned scores and 12 use score-independent random selection. The official GSM8K test split was not read. Early preparation accessed the full train table, so secondary83 is a coverage-shift set rather than an untouched benchmark.

Source groups contain different MetaMathQA types, including answer augmentation, rephrasing, FOBAR, and self-verification variants. Matching type within each source controls this variation while allowing the records' content to differ. The mapping identifies lexical source relationships; upstream generation lineage, semantic overlap, and pretraining contamination remain outside this check.

B. Exposure and Optimization

We format each prompt with the instruction and response markers and truncate downstream sequences to 1,024 tokens. Let c_i denote nonpadding input tokens and u_i supervised tokens after the causal shift. Selection satisfies

$$\sum_{i \in S} c_i \leq 2{,}000{,}000. \tag{1}$$

Records are indivisible; a unit that does not fit is skipped while scanning continues. Within-source manifests nearly ex-

haust the ceiling, whereas grouping-policy manifests use 1.63–1.66M tokens because their support differs.

Each fit starts from the frozen Qwen2.5-1.5B base with fresh q/v LoRA (rank 8, alpha 16, dropout 0) and AdamW states. AdamW uses learning rate 2×10^{-4}, betas $(0.9, 0.999)$, epsilon 10^{-8}, and weight decay 0.01, with no warmup or clipping. Records are traversed once in a seed-specific order. Let $U_k = \sum_{i=1}^{k} u_i$, $T = U_N$, and $K = 1{,}024$. For $j < K$, boundary b_j minimizes $|KU_{b_j} - jT|$ over $b_{j-1} + 1 \le b_j \le N - (K - j)$, with earlier boundaries breaking ties; $b_0 = 0$ and $b_K = N$. Each block's summed answer-token loss is divided by its supervised-token total before one update. Thus total exposure and updates match while content, order, block membership, and optimizer trajectories can differ.

C. Ranking Within Fixed Allocations

An exact cell is $(\text{source ID}, \text{type}, c_i, u_i)$. Sorted cells are shuffled by construction seed. For a cell containing m records of input cost c, with remaining budget R, both arms select

$$q = \min\left(\lfloor m/2 \rfloor, \lfloor R/c \rfloor\right). \tag{2}$$

These cell quotas induce source quota $Q_s = \sum_{h:\,\text{source}(h)=s} q_h$. Quotas are fixed between paired arms within each construction and can change across construction seeds. The ranked arm takes the q highest-scoring records, breaking ties by row ID. The random arm samples q uniformly without replacement from the entire cell using a score-independent cell seed, so the arms can overlap. Singleton cells receive zero quota. Eligible selections match source identities, source quotas, types, record counts, and both token totals (Fig. 1). Figure 2 separates this controlled question from the broader global comparison.

Construction seeds 42, 43, and 44 yield source coverage of 63.36%, 62.69%, and 63.17%, respectively. Each construction is crossed with training seeds 42, 43, and 44. The primary estimand is the equal-weight mean over this observed 3×3 grid, conditional on the shared scorer, manifests, and fitted models.

D. Grouping and Global Comparisons

A separate experiment compares real source labels with permuted labels preserving the group-size distribution. Shared cells are (type, c_i, u_i). Common cells are shuffled; each arm independently shuffles unused groups before allocating the same number. Each allocated group contributes its highest-scoring record in that cell, with row-ID tie-breaking, and is then permanently removed. Cell quotas are bounded by both arms' available groups and remaining budget.

Real labels cover 6,574–6,589 true sources (95.58–95.80%), compared with 4,370–4,386 (63.54–63.77%) for permuted labels. Paired arms exactly match input tokens, supervised tokens, and 1,024 updates. Construction and permutation seeds are bound as c42/p1042, c43/p1043, and c44/p1044. Changing group–cell associations and concrete allocations makes this a joint grouping-policy comparison rather than an isolated causal effect of coverage.

Global LESS and global row random instead allow source allocations to vary. Each uses selection seed42 and three training seeds, the same input ceiling, and 1,024 updates. Their record counts and supervised-token totals differ, so they estimate overall policy differences.

IV. Scoring and Evaluation

A. LESS Adaptation and Initialization Diagnostic

The LESS adaptation follows author-code commit `8abf9628` [4]. A shared 5% warmup (11,177 records, seed42) runs for four epochs with q/k/v/o LoRA (rank 128, alpha 512, dropout 0.1) and one checkpoint per epoch. Warmup and candidate/probe scoring use 2,048 tokens; downstream training and cell costs use the 1,024-token view. Warmup uses learning rate 2×10^{-5}, effective batch32, clipping 1.0, zero weight decay, a 0.03 warmup ratio, and linear scheduling.

At each checkpoint, candidate gradients and saved Adam moments define update directions, while probe256 supplies ordinary loss gradients. TRAK 0.1.3 uses an 8,192-dimensional Rademacher projection with seed0. Projected features are normalized, candidate–probe cosines are averaged over probe questions, and checkpoint scores are weighted by epoch-average learning rates. All LESS selections use one shared batch16 scoring realization, and downstream fits start from the base model. On the checked shard, batch16 failed all four numerical-equivalence gates against the earlier batch8 scoring implementation. The reported results apply to this LESS adaptation.

The initialization-gradient diagnostic computes 4,096-dimensional CountSketch features from response-token mean-loss gradients at q/v LoRA initialization. Ranking uses cosine similarity to the raw mean of probe features, without warmup or candidate Adam update directions. It provides a separate test of initialization-based relevance under the same allocation controls.

B. Execution Grid and Matching Metrics

The study contains 51 downstream models. The initialization-gradient and grouping comparisons each use 18 fits. The within-source LESS comparison uses nine LESS fits paired with the nine exactly replayed random models from the initialization-gradient comparison. Global LESS and row random use three fits each.

Initialization-gradient and grouping evaluations use batch1 greedy decoding on dev256, with at most 512 new tokens. All LESS comparisons use batch8 predictions, left padding, attention masks, and the same greedy cap, including reevaluation of the reused random models. Scores are compared within each decoding configuration because full-question token equivalence between batch1 and batch8 is unestablished.

Strict matching parses the text after the last case-insensitive `####` or `The answer is:` marker as a signed decimal, allowing valid thousands separators, surrounding whitespace, an optional dollar prefix, and a final period. Extra prose or units are rejected. Trailing-number matching extracts the

TABLE I
SELECTION EXPOSURE. WITHIN-SOURCE ROWS APPLY TO BOTH LESS AND RANDOM; EVERY FIT USES 1,024 UPDATES.

Manifest	Records	Input tokens	Supervised tokens
Within c42	9,718	1,999,998	1,233,666
Within c43	9,678	1,999,991	1,235,783
Within c44	9,672	1,999,926	1,236,156
Global LESS	10,196	1,999,953	1,203,674
Global random	7,872	1,999,967	1,321,452

last numeric token without requiring a marker. Both compare values exactly using decimal arithmetic. Rejected parses and capped generations remain in the denominator, so matching measures numerical reference agreement.

The LESS evaluation comprises 75 model–set evaluations. Nine matched-random dev evaluations were added to the initial 66 after aggregate results were available. A plan froze these pairs before supplemental prediction, without retraining or hyperparameter changes, making the within-source dev comparison retrospective. Evaluations share questions, which are resampled jointly in the uncertainty analysis.

C. Uncertainty

We take 10,000 paired question-bootstrap draws, seed42, with shared question indices across every model pair [21]. The 2.5th and 97.5th percentiles form conditional intervals. Equivalently, paired differences are first averaged per question and those question averages are resampled. These intervals quantify question-sampling uncertainty conditional on the shared scorer and fitted models; scorer, construction, and retraining uncertainty are excluded, and no multiple-comparison adjustment is applied [22], [23]. Sample SD with $\mathrm{ddof} = 1$ describes variation across fitted pairs. Single-seed omission checks describe sensitivity within the observed grid, rather than uncertainty for a new scorer or a complete rerun.

V. RESULTS AND ANALYSIS

A. Matched Exposure and Ranking Value

Table I shows why allocation and exposure controls matter. Within-source LESS and random match both input and supervision. Global LESS uses 1,203,674 supervised tokens compared with 1,321,452 for global random, an 8.91% reduction despite nearly identical input totals. The paired design removes this exposure difference from the within-source comparison.

On the observed 3×3 dev256 grid, within-source LESS achieves 79.731% mean strict reference matching versus 78.429% for matched random selection, a 1.66% relative increase (Table II). The matched source quotas and token totals make this a comparison of record choices within the same allocation. Six of the nine fitted-pair differences are positive and three negative (Fig. 3). All three construction means and all three training-seed means are positive.

The positive mean difference persists after omitting any one construction seed or training seed (Fig. 4). The conditional

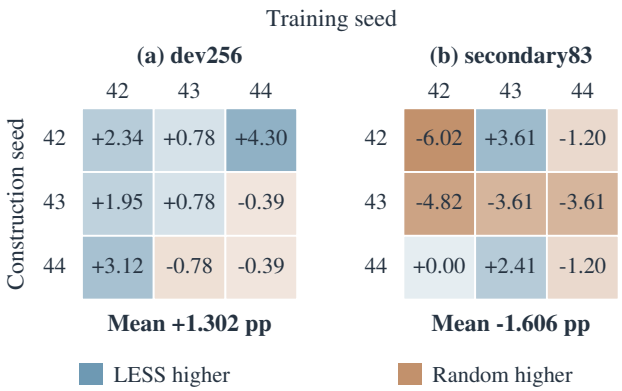


Fig. 3. Paired strict reference-match differences for the LESS adaptation minus matched random, in absolute percentage points (pp), inference batch8. Rows are construction seeds; columns are training seeds. The retrospective dev256 comparison and coverage-shift diagnostic share a zero-centered scale. Secondary83 reference validity is examined in Section V-C.

95% interval for the absolute difference is $[0.000, 2.604]$ percentage points. Under the marker-free trailing-number parser, the same predictions yield 79.818% matching for LESS and 78.429% for random, a 1.77% relative increase. Two additional LESS numerical matches account for the change, with a conditional absolute-difference interval of $[0.043, 2.734]$ percentage points.

B. Auxiliary Ranking and Coverage Diagnostics

Table III separates initialization-gradient ranking from the grouping-policy comparison under batch1 decoding. Initialization-gradient ranking increases mean scoring cosine in every construction, while downstream differences change sign across construction means. Individual correct-count differences range from -14 to $+9$. The diagnostic distinguishes improvement in the selection score from improvement in downstream reference matching.

Real-source grouping covers approximately 96% of sources compared with 64% for permuted grouping and slightly increases mean cosine. Mean strict matching is 79.427% versus 80.425%, respectively, a relative difference of -1.24%, with negative differences in all three construction means. Source coverage and downstream matching therefore capture distinct properties of the selected data. This comparison evaluates the joint grouping policy, including its changed allocations and group–cell associations.

C. Coverage-Shift Evaluation Diagnostic

Secondary83 tests reference matching on the mapping-selected complement. Within-source LESS and random score 5.756% and 7.363%, respectively (Table II); their nine correct-count differences are $-5, +3, -1, -4, -3, -3, 0, +2, -1$ (Fig. 3). The mean difference remains negative under every single-seed omission (Fig. 4). Global differences are negative on dev and positive on secondary, with intervals crossing zero. These global comparisons measure overall policy differences involving allocation, content, and supervision.

TABLE II
STRICT MATCHING FOR THE LESS ADAPTATION, INFERENCE BATCH8. DIFFERENCES AND SD ARE ABSOLUTE PERCENTAGE POINTS (PP); DIFFERENCES ARE LESS MINUS RANDOM. INTERVALS RESAMPLE SHARED QUESTIONS CONDITIONAL ON FITTED MODELS. † RETROSPECTIVE MATCHED-DEV COMPARISON.

Set	Scope	Pairs	LESS (%)	Random (%)	Δ (pp)	SD (pp)	Conditional 95% interval
dev256†	Within source	9	79.731	78.429	+1.302	1.747	[0.000, 2.604]
dev256	Global	3	78.385	79.818	−1.432	2.542	[−4.818, 1.823]
secondary83	Within source	9	5.756	7.363	−1.606	3.244	[−3.481, 0.000]
secondary83	Global	3	8.434	8.032	+0.402	3.032	[−4.016, 4.819]

TABLE III
AUXILIARY DEV256 STRICT MATCHING, INFERENCE BATCH1. EACH COMPARISON HAS NINE CONSTRUCTION–TRAINING PAIRS. DIFFERENCES AND SD ARE ABSOLUTE PERCENTAGE POINTS (PP).

Comparison	Left (%)	Right (%)	Δ (pp)	SD (pp)
Init. gradient − random	79.123	78.559	+0.564	3.039
Real − permuted	79.427	80.425	−0.998	1.915

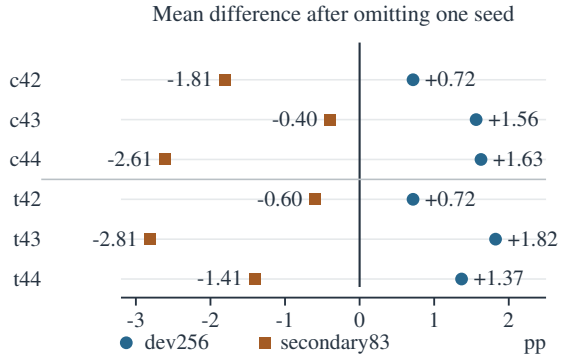


Fig. 4. Single-seed omission sensitivity, LESS adaptation minus matched random, in absolute percentage points (pp). Each row omits one construction or training seed from the 3×3 grid. Full-grid means are $+1.302$ pp on dev256 and -1.606 pp on secondary83; both retain their sign under every omission. These are descriptive grid checks.

Secondary means of 5.8–8.4% are much lower than dev means of 78.4–79.8%. Replaying frozen parsers on the 1,494 secondary predictions from the 18 within-source models finds 12 strict rejections, seven cap-reaching outputs, and no empty outputs. Relaxed parsing adds one match. Of the predictions passing strict parsing, 697 LESS and 687 random outputs disagree numerically with their reference. Formatting and truncation cannot explain the overall gap.

A retrospective qualitative sample exposed reference contradictions. In one case, the question specifies 20 trucks and five times as many tanks. Both models output 120, consistent with $20 + 5 \times 20$, but the reference's final marker says 140 despite computing 120 in its body. Another case contains both a flawed reference and incorrect model reasoning. A reference defect therefore does not certify a model answer.

An AI-assisted census covered all 83 references in three disjoint shards with model outputs and correctness withheld. Rechecking all 18 initial clear-error candidates yielded 16 clear errors, 26 ambiguous or underspecified cases, 40 reference-consistent cases, and one unresolved case. Auditors knew of prior issues; ambiguous and consistent cases were not all independently rechecked, and no human adjudication was performed. The retained summary contains counts and examples but lacks the referenced item-level shards, audit plan, and auditor model/version. Query, reference, and ID equality checks authenticate transfer from the frozen local GSM8K train source, not the upstream distribution.

We retain original references, scores, and denominators and compute no corrected accuracy or revised ranking. The audit localizes some disagreements but does not explain the overall low score or estimate reference-error prevalence in GSM8K. Secondary83 cannot supply independent confirmation of a mathematical-ability difference.

VI. DISCUSSION AND LIMITATIONS

Implications for data curation. Fixed allocations make relevance-guided substitution measurable, while global comparisons evaluate allocation and ranking jointly. Coverage describes the selected data's structure; downstream evaluation measures its utility. Shared instruction-data workflows can preserve these distinctions by recording source IDs, quotas, input and supervised-token totals, and update counts alongside ranking results.

Scope. The study evaluates English mathematical instructions with a 1.5B model, LoRA, and a roughly 2M-token budget. Exact matching applies to cells with alternatives and sufficient remaining budget. The approximately 63% source coverage describes selected manifests; exact-cell record support and replacement intensity are not quantified. Matched totals permit different training trajectories, and the reported intervals describe question-sampling uncertainty for the shared scorer and fitted models.

Evaluation and provenance. Repeated use of dev256 and the retrospective matched-random comparison delimit the present evidence to an observed development-set result. Broader ability claims require independent evaluation. Secondary83 serves as a coverage-shift diagnostic, with the reference-validity findings reported in Section V-C. Frozen data and model versions were acquired through public mirrors with incomplete independent verification against official hosts and licenses.

Cost and reproducibility. Warmup took 5,391.253 seconds for 1,400 updates; 516 scoring workers recorded 620,701.535 seconds in aggregate. Overlapping durations and incomplete

stage timings preclude end-to-end elapsed-time or GPU-cost comparisons. The LESS analysis uses 10,377 saved outputs from 75 evaluations. Historical batch1 results retain per-pair counts but lack per-question predictions and scoring records. Member-level selections, model weights, training inputs, and gradient matrices needed to reconstruct selection, training, and inference were not retained.

VII. Conclusion

We introduced a source-controlled protocol for evaluating instruction ranking under fixed source allocations and matched training exposure. Pairing ranked and random records within exact source, type, and token-length cells separates within-source selection from changes in which problems receive training effort. In the retrospective MetaMathQA–Qwen2.5-1.5B dev256 comparison, LESS-based ranking achieves a 1.66% relative increase in mean strict reference matching over matched random selection, with positive averages across all three constructions. The global comparisons further demonstrate that nearly identical input-token budgets can conceal different amounts of supervised training. Together, the protocol and results show how source quotas and realized supervision make instruction-selection comparisons interpretable.

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